



HIV-RTx Mutation Profiling

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Introduction



Diagnosis

- HIV Subtype
- Viral Load
- Comorbidities
- CD4 Cell count
- ART naive?



Regimen





- First Line Regimen

National treatment guidelines

Effectiveness against the most common HIV strains

Low risk of side effects

Convenience (often one pill once daily)

- Second Line Regimen

Virologic failure

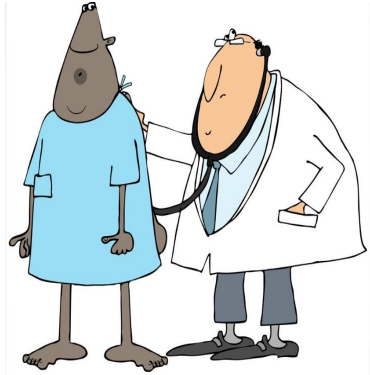
Severe side effects or toxicity

Significant drug interactions or contraindications

Drug resistance

1. NRTIs
2. NNRTIs
3. Integrase -I
4. Protease -I

Introduction ... diversion



Diagnosis



- HIV Subtype
- Viral Load
- Comorbidities
- CD4 Cell count
- ART naive?
- RTx profile



- Routine genotypic resistance testing is available (expensive, chosen few)
- TDRMs are harder to trace - costly to treat during this day and era (Need intervention)

First line (NRTIs) - RTx profile



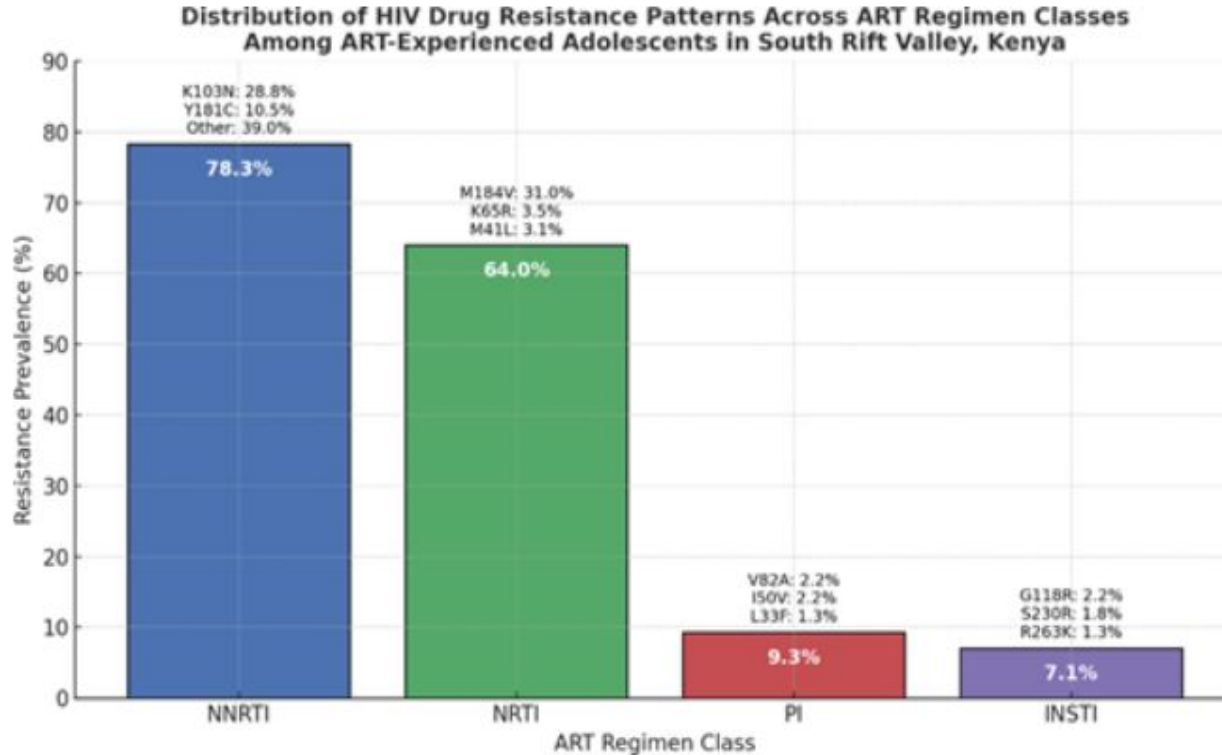
Regimen



Objectives

- To identify RT mutations associated with NRTI resistance using the SDRDB
- To develop ML model to predict NRTI resistance from HIV RT mutation profiles
- To compare predictive performance and interpretability of ML algorithms for resistance prediction
- To discuss implications for regimen selection in Kenya.

Drug Selection Criteria



- 3TC
- TDF
- ABC
- AZT

Mutation (Feature Selection) - 42

DRUG	Isolates	Mutations
3TC	2359	42
TDF	2012	42
ABC	2231	42
AZT	2381	42

Model Evaluation

Priority - True Negatives & True Positives

- More cost for false negatives than for false positives
- More than accurate

$$\text{Precision} = \frac{\text{True Positives (TP)}}{\text{True Positives (TP)} + \text{False Positives (FP)}}$$

$$\text{Recall} = \frac{\text{True Positives (TP)}}{\text{True Positives (TP)} + \text{False Negatives (FN)}}$$

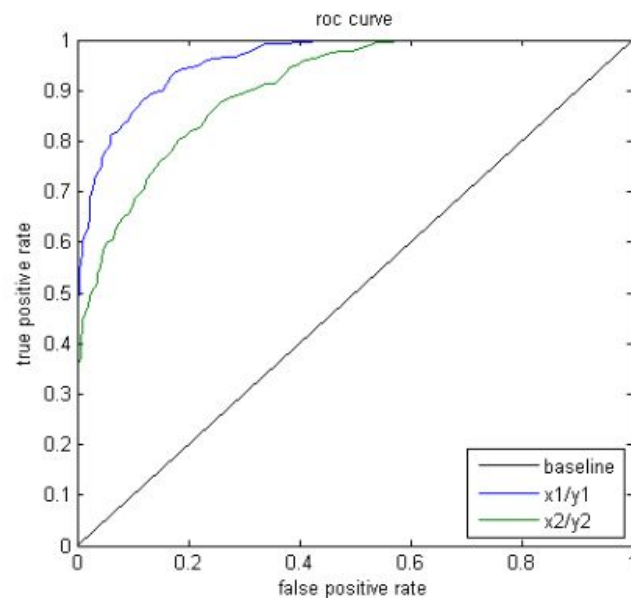
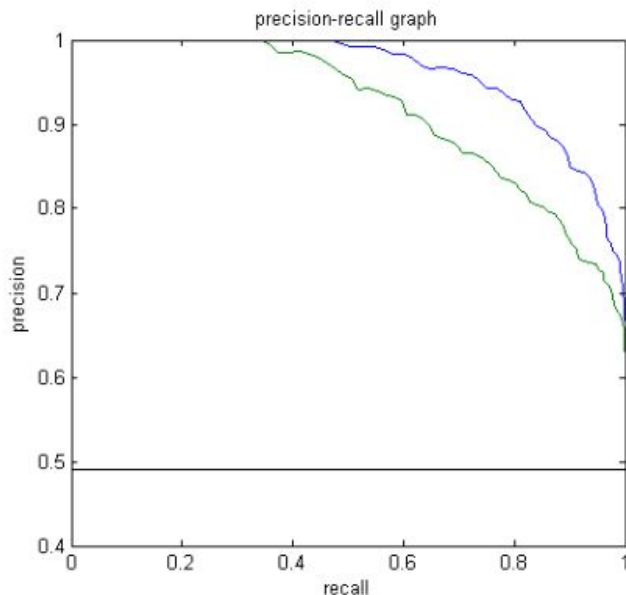
$$\frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

Priority - True Negatives & True Positives

- Consequences of different types of errors are not equal (ROC and PR curves were important)

$$TPR = \frac{TP}{TP+FN}$$

$$FPR = \frac{FP}{FP+TN}$$



Individual Models Metrics Evaluation Report

Metric	Logistic Regression	Random Forest	Support Vector Machine	Extreme Gradient Boost
Classification Report	✕	✕	✕	✕
Confusion Matrix	✕	✕	✕	✕
Check for Overfitting	✕	✕	✕	✕
ROC-AUC curve plotting	✕	✕	✕	✕
Precision Recall curve	✕	✕	✕	✕

Model Training

DRUG	Isolates	Mutations
<u>3TC</u>	2359	42
<u>IDF</u>	2012	42
<u>ABC</u>	2231	42
<u>AZI</u>	2381	42

Summary

Drug	Best Model
3TC	Extreme Gradient Boost (XGBoost)
TDF	Logistic Regression (LR)
AZT	Random Forest (RF)
ABC	Extreme Gradient Boost (XGBoost)

Integration and Deployment (3TC)

```
import pandas as pd

def predict_3TC(mutations, model, feature_columns):

    # Create empty mutation vector
    patient = pd.DataFrame( 0, index=[0], columns=feature_columns )

    # Encode mutations
    for mut in mutations:
        if mut in patient.columns:
            patient.loc[0, mut] = 1
        else:
            print(f"{mut} not found in training features.")

    # Predict
    pred = model.predict(patient)[0]
    prob = model.predict_proba(patient)[0,1]

    print("Mutations:", mutations)
    print("Prediction:",
          "Resistant" if pred else "Susceptible")
    print("Resistance probability:",
          round(prob,3))

    return patient

predict_3TC(test_mutation, best_models["3TC"], feature_columns)
```

Prediction: Susceptible
Resistance probability: 0.419

Integration and Deployment (TDF)

```
import pandas as pd

def predict_TDF(mutations, model, feature_columns):

    # Create empty mutation vector
    patient = pd.DataFrame( 0, index=[0], columns=feature_columns )

    # Encode mutations
    for mut in mutations:
        if mut in patient.columns:
            patient.loc[0, mut] = 1
        else:
            print(f"{mut} not found in training features.")

    # Predict
    pred = model.predict(patient) [0]
    prob = model.predict_proba(patient) [0,1]

    print("Mutations:", mutations)
    print("Prediction:",
          "Resistant" if pred else "Susceptible")
    print("Resistance probability:", round(prob,3))

    return patient

predict_TDF(test_mutation, best_models["TDF"], feature_columns)
```

Prediction: Resistant
Resistance probability: 0.504

Integration and Deployment (AZT)

```
import pandas as pd

def predict_AZT(mutations, model, feature_columns):

    # Create empty mutation vector
    patient = pd.DataFrame( 0, index=[0], columns=feature_columns )

    # Encode mutations
    for mut in mutations:
        if mut in patient.columns:
            patient.loc[0, mut] = 1
        else:
            print(f"{mut} not found in training features.")

    # Predict
    pred = model.predict(patient)[0]
    prob = model.predict_proba(patient)[0,1]

    print("Mutations:", mutations)
    print("Prediction:", "Resistant" if pred else "Susceptible")
    print("Resistance probability:", round(prob,3))

    return patient

predict_AZT(test_mutation, best_models["AZT"], feature_columns)
```

Prediction: Resistant
Resistance probability: 0.87

Integration and Deployment (ABC)

```
import pandas as pd

def predict_ABC(mutations, model, feature_columns):

    # Create empty mutation vector
    patient = pd.DataFrame( 0, index=[0], columns=feature_columns )

    # Encode mutations
    for mut in mutations:
        if mut in patient.columns:
            patient.loc[0, mut] = 1
        else:
            print(f"{mut} not found in training features.")

    # Predict
    pred = model.predict(patient)[0]
    prob = model.predict_proba(patient)[0,1]

    print("Mutations:", mutations)
    print("Prediction:", "Resistant" if pred else "Susceptible")
    print("Resistance probability:", round(prob,3))

    return patient

predict_ABC(test_mutation, best_models["ABC"], feature_columns)
```

Prediction: Resistant
Resistance probability: 0.791

'DawaFit' Container



DAWAFIT

	Drug	Prediction	Resistance Probability
0	3TC	Susceptible	0.419
1	AZT	Resistant	0.870
2	TDF	Resistant	0.504
3	ABC	Resistant	0.791

Recommended drugs: 3TC

Avoid: AZT, TDF, ABC

